PARTH JADHAV

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PROFESSIONAL SUMMARY

M.Sc. Artificial Intelligence and Machine Learning student with hands-on experience building AI/ML, multimodal deep learning, RAG and LLM-based applications. **Engineered a scalable multi modal deep learning system** for e commerce pricing that **placed 38th out of 9,000+ teams** in the Amazon ML Challenge 2025. Delivered an AI powered learning path generator that **attained 100% anti hallucination accuracy** on test dataset with retrieval augmented generation. Strong analytical mindset, collaborative approach, and proven ability to translate complex data into deployable models.

EDUCATIONS

Vellore Institute of Technology | Vellore, India — 2025 - 2027
Master of Science in Artificial Intelligence and Machine Learning — *CGPA : 8.85/10*

Vivekanand College | Kolhapur, India — May 2023
Bachelor of Science in Computer Science — *CGPA : 8.63/10.0*

SKILLS

Databases :	SQL, MySQL
Frameworks & Libraries :	PyTorch, TensorFlow, Scikit learn, Pandas, NumPy
NLP & Gen AI :	FAISS, RAG, BERT, Sentence Transformers
Programming Languages :	Python, Java
AI/ML :	Machine Learning, Model Training, Model Evaluation, Large Language Models (LLMs), Data Science, Transformer, Supervised & Unsupervised Learning
Tools & Platforms :	Flask, VS Code

PROJECTS

RAG-Based Machine Learning Study Assistant

- Developed a domain-specific educational AI assistant utilizing Retrieval-Augmented Generation (RAG) to deliver context-aware answers from academic texts. Built a robust, multi-modal data ingestion pipeline with PyMuPDF for native text and PyTesseract for OCR, minimizing data loss from complex PDF layouts and diagrams.
- Implemented an in-memory dense vector semantic search system using the all-MiniLM-L6-v2 embedding model and a FAISS IndexFlatIP vector store. Orchestrated natural language synthesis with the Qwen2.5-7B-Instruct LLM via Hugging Face's serverless Inference API. Designed strict system prompts and fine-tuned hyperparameters to ensure rigorous adherence to injected context and prevent model hallucinations. Deployed the end-to-end machine learning pipeline as a RESTful web application using the Flask framework. Significantly improving evaluation accuracy and turnaround time for academic reporting.

Technologies / Tools Used : *Python, MySQL, Flask, PyMuPDF, PyTesseract, FAISS, Sentence Transformers, Qwen2.5-7B, Hugging Face, RAG*

Multi Modal Product Pricing Predictor — Dec 2025 - Dec 2025

- **Secured All India Rank 38** (out of 9,000+ teams) by engineering a scalable, multi modal deep learning system to predict e commerce product prices.
- **Developed a dual encoder architecture** integrating a **BERT** based model for processing textual product descriptions and a **Vision Transformer (ViT)** for extracting image features.
- **Built a custom feature fusion layer** with a regression head to combine text and image modalities, bridging the gap between academic theory and industry grade scalability.
- **Optimized model convergence and performance** by implementing a custom SMAPE loss function, AdamW optimizer.

Technologies / Tools Used : *PyTorch, Python, BERT, NumPy, Pandas, Vision Transformer(ViT)*

Personalized AI Powered Learning Path Generator

Developer

- **Researched latest AI trends** and collaborated to design a full stack AI web application generating goal driven learning roadmaps using automated LLM ontology extraction.
- **Evaluated and optimized a Retrieval Augmented Generation (RAG)** pipeline using ChromaDB and Sentence Transformers to restrict model responses to verified textbook content, achieving **100% anti hallucination accuracy**.
- **Developed an Explainable AI (XAI)** module with MySQL caching to generate natural language justifications, successfully optimizing the system to reduce repeat API calls to zero.
- **Integrated AI solutions** into a dual interface dashboard, allowing dynamic PDF uploads, vector monitoring, and real time query testing.

Technologies / Tools Used : *SentenceTransformer, LLM, RAG, XAI, Python, Flask*

WORK EXPERIENCES

Android Developer Intern | Internship — Jun 2023 - Dec 2023
Dropout Technologies LLP — *On-site • Pune, India*

- Worked on application modules involving data handling and API integration.
- Used Kotlin and SQLite for structured data storage and retrieval. •
- Integrated APIs to enable wireless device communication.
- Gained hands-on experience with debugging, testing, and performance optimization.

Technologies / Skills Used : *Kotlin, SQLite*

CERTIFICATIONS

• Artificial Intelligence with Python: Friends Union Energising Lives (FUEL): Pune, India
• Introduction to Cybersecurity Tools & Cyberattacks: Coursera (Online)